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Md Saif Ahmed

Backend Engineer | Go · Python · PostgreSQL | Competitive Programmer

Backend engineer (CSE, MIST, graduating 2027), open to remote roles. **Sole engineer of Hasnahena**, a live finance platform (Go, PostgreSQL) used by real members since Aug 2026, and builder of a team Q&A app's FastAPI backend. Competitive programmer (**38th at ICPC Asia Dhaka Regional 2025**, 1700+ problems solved).

Live	~180/s	269	1700+
solo-built finance platform, since Aug 2026	postings in load test, reconciled exactly	Go tests, incl. race-detector runs	problems solved • CF Specialist

PROJECTS

Hasnahena — Interest-Free Cooperative Finance Platform (Solo, in production) hasnahenafoundation.com | GitHub

Go 1.25 (chi, pgx), PostgreSQL (Neon), Next.js 16, TypeScript, Prisma, Docker, GitHub Actions, Cloudflare, Render

- Designed, built and operate a platform digitising a Jashore cooperative's passbook, **live with real members since Aug 2026**: deposits, interest-free loans with guarantors, donations and disputes across three roles.
- Go REST API (60+ endpoints) owns every balance-changing write, each in **one PostgreSQL transaction** with `FOR UPDATE` row locks and advisory locks; integer money (never float) guarded by CHECK constraints and a **trigger-enforced append-only audit log**; RBAC, JWT with server-side revocation, rate limiting.
- pgxpool tuning, HTTP timeouts, structured logging (slog) and graceful shutdown that closes SSE streams before draining in-flight requests; health and readiness probes.
- **269 Go tests**: integration tests on real Postgres under `-race`, 400k-iteration money property tests and a load test sustaining **~180 postings/s** on contended accounts that reconciles exactly.
- Operate it with GitHub Actions CI, Docker images on GHCR, encrypted daily backups to Cloudflare R2 with a CI restore drill (**RPO 24 h, RTO < 1 h**), Sentry, Prometheus and UptimeRobot; documented in **45 ADRs** and runbooks.

QuestBoard — Gamified Q&A Platform for STEM Students (Team) GitHub

Python, FastAPI, SQLAlchemy, PostgreSQL (Supabase), Anthropic Claude API, Codeforces API, pytest, Flutter, Render

- Students post "quests" with point bounties. **Wrote all of the current server code**: FastAPI, 7 routers, ~40 endpoints; Postgres schema with race-safe voting (UNIQUE), trigram search, RLS, and a point ledger moving points **atomically in one transaction**.
- **Anthropic Claude API** hints, rate-limited and refunded on failure; Codeforces API daily challenges and handle verification; **70 pytest tests** run in CI against a real Postgres; deployed on Render.

LocalFix — Civic Engagement Platform (Team) GitHub

React (Vite), Node.js, Express.js, Oracle Database (PL/SQL), JWT, bcrypt, Helmet, Mapbox GL

- Complaint management system where citizens report civic issues (roads, waste) to authorities; role-based access control and optimized Oracle SQL queries **cut data retrieval latency by 40%**.
- Wrote ~95% of the Express API (Helmet, rate limiting, validation, JWT) and the Oracle schema: **14 triggers, 3 stored procedures, 3 views, 27 indexes**; plus a large share of the React client.

HallBridge — Digital Hall Management System (Team) GitHub

Next.js 16, React 19, TypeScript, MongoDB (Mongoose), Tailwind CSS, Supabase, Google OAuth 2.0

- Seat allocation, billing and notices for **500+ residents**, with student, staff and admin dashboards and a predictive meal planner that brought food wastage to near-zero.
- Wrote ~64% of the API and ~81% of the data models; **Google OAuth 2.0** sign-in with a CSRF state cookie, password reset with expiring tokens, and compound indexes with lean, projected reads.

MCC Website — MIST Computer Club Platform (Team, in production) GitHub

Next.js, Bun, Hono, PostgreSQL, JWT, Supabase, Docker Compose, GitHub Actions

- The club's live platform (Next.js, Bun/Hono API, Discord bot; Docker Compose). Built the **alumni directory end to end**: public listing, JWT-protected admin CRUD, Postgres migrations, Next.js pages; landing-page admin.
- Hardened contest-report rooms: input validation, idempotent adds and a **broken access control fix**, so read endpoints require an admin instead of any signed-in user.

BruteBuddy — Automated Security Testing Tool GitHub

Python, requests, ThreadPoolExecutor, CLI

- CLI for **automated fuzzing** and brute-force testing of login endpoints: multi-field and generated payloads, custom headers and cookies, automatic re-authentication; parallel requests run **3× faster**.

GyanTori — Peer-to-Peer Book Sharing (Team) GitHub

Java, JavaFX (FXML), Maven, MongoDB

- Desktop virtual library for **500+ students** to lend, borrow and catalog books, with a MongoDB schema for inventory relationships and transaction history.

EXPERIENCE

Industrial Trainee, JB Connect Ltd (JBC) 2026 | 4 weeks

Industry sessions on DevOps, backend engineering and AI/ML; tasks with the company's teams

- Team AI content-generation app (FastAPI, React): set up the backend's router → service → repository layers, env-driven settings, error-to-HTTP handlers, Alembic and Docker dev setup.
- Built the `POST /api/v1/generate` endpoint with a mock AI client behind an injectable FastAPI dependency; shipped through pull requests.

TECHNICAL SKILLS

Languages	Go (Golang), Python, TypeScript, JavaScript, SQL, C++, Java, Bash
Backend	Go (chi, pgx, slog), FastAPI, Pydantic, Node.js, Express.js, Hono, Bun, REST APIs, SSE, JWT, OAuth 2.0, RBAC
Databases	PostgreSQL (transactions, row locks, constraints, triggers, indexing), Prisma, SQLAlchemy, Alembic, MongoDB, Oracle PL/SQL
DevOps & Cloud	Docker, GitHub Actions, Linux, Render, Vercel, Neon, Cloudflare (R2, Workers), Sentry, Prometheus, UptimeRobot
Testing	Go testing (unit, integration, `-race`, property, load), pytest, Playwright, Vitest, golangci-lint, Ruff
Frontend	React, Next.js, Tailwind CSS, Flutter
AI / LLM	Anthropic Claude API, OpenAI-compatible APIs, mockable AI clients for tests
Core CS	Data structures & algorithms, OOP, concurrency control, system design, OWASP Top 10, secure coding

EDUCATION

B.Sc. in Computer Science & Engineering 2023 – 2027 (Expected)

Military Institute of Science and Technology (MIST), Dhaka | *CGPA* **3.58 / 4.00** *(up to Level-3, Term-2)*

Higher Secondary Certificate (HSC) | Govt. M. M. College, Jashore | GPA 5.00 / 5.00 2022

TOOLS & SCRIPTS

- **Rank-Raven** *(Python, Selenium, BeautifulSoup, ReportLab)*: Codeforces activity reporter that generates styled PDF reports.
- **upsolve_tracker** *(Apps Script, Sheets, Codeforces API)*: schedules near-miss problems, times solves, detects AC automatically.
- **Code-Library** *(C++, LaTeX, Python)*: MIST_EagleForces ICPC team notebook with a script that builds the PDF.
- **Faculty-Evaluation** *(Python, Selenium)*: automates the university's faculty evaluation form.

COMPETITIVE PROGRAMMING

Codeforces Specialist (max 1514) | **CodeChef** 3-Star (max 1722) | **1700+** problems solved (CF, CodeChef, LeetCode, CSES, AtCoder)

ICPC Asia Dhaka Regional 2025 *(univ. teams)*	**38th**	ICPC Asia Dhaka Regional 2024 *(univ. teams)*	**117th**
ICPC Dhaka 2025 Online Preliminary	**290th** / 1842	ICPC Dhaka 2024 Online Preliminary	**90th** / 2490
DUET IUPC 2025	**16th** / 137	NSUCEC Cybernauts IUPC 2026	**16th** / 159
CUET IUPC 2025	**28th** / 131	DUET IUPC 2026	**24th** / 139
UIU IUPC 2025	**43rd** / 161	NDUB IUPC 2026	**32nd** / 85
AUST IUPC 2025	**48th** / 129	SUST CSE Carnival IUPC 2026	**44th** / 137
MU CSE Fest IUPC 2025	**49th** / 99	IUT ICT Fest IUPC 2026	**46th** / 111
Uttara University IUPC 2025	**54th** / 122	KUET IUPC 2025	**89th** / 161
BUP IAUPC 2025	**Champion** / 28	Meta Hacker Cup 2024	**Round 2**

LEADERSHIP & TEACHING

President, MIST Computer Club Jun 2026 – Present

Vice President (May 2025 – Jun 2026) | *Executive Member (May 2024 – May 2025)*

- Lead club strategy and mentor **15+ core members** delivering **10+ technical events a year**; direct ICPC preparation camps and workshops reaching **200+ students**.
- As Vice President, coordinated programming workshops, contest initiatives and training for junior members.

Competitive Programming Instructor, MIST Computer Club May 2024 – Present

- Mentor **50+ junior students** in algorithms and contest strategy with an **80% retention rate**; curated **20+ problem sets** and mock contests, helping teams advance in ICPC preliminary rounds.

Web Security Instructor, MIST Cyber Security Club May 2024 – Jun 2026

- Ran hands-on **OWASP Top 10** workshops with real-world exploitation scenarios; trained **30+ students** in penetration testing and secure coding. Also Associative Member (May 2024 – Jun 2025).

Executive Director, Mentor Committee | coordinated security mentoring initiatives and learning tracks Jun 2025 – Jun 2026

Hackathons: MIST Intra Hackathon 2025 | BUP CSE Fest 2026 Hackathon (onsite)

CYBERSECURITY & CTFS

Cyber Drill 2025 (WTISD Hackathon)	**5th**	Bugcrowd USA CTF 2025	**34th global**
SMP CTF 2024	**13th**	BUET CTF 2024 (final round)	**18th**
Cyber Raid 2025 (final round)	**18th**	DIU Cybercon CTF 2025 (final round)	**18th**
BCS ICT Fest CTF 2025	**18th**	BUP CTF 2025	**18th**

Hands-on practice: PortSwigger Academy **70+ labs** (10+ categories) | PicoCTF **67+** | TryHackMe **29 rooms** | OverTheWire Natas
Training: Hands-on Ethical Hacking and Security Analysis (MIST Cyber Security Club) | eShikhon Ethical Hacking Live Course